

60016 OPERATIONS RESEARCH

Sensitivity Analysis

Lecture 08

Program

Last lecture

- ▶ Duality

This lecture

- ▶ Value function
- ▶ Shadow prices

In a nutshell: how does solution of LP depend on params?

Example 1 (perturbed)

Assume p_1 , the availability of machine X, is not precisely known.

$$\begin{array}{ll} \max & y = x_1 + x_2 \quad : \text{objective function} \\ \text{s.t.} & 2x_1 + x_2 \leq p_1 \quad : \text{constraint on availability of machine X} \\ & x_1 + 3x_2 \leq 18 \quad : \text{constraint on availability of machine Y} \\ & x_1 \leq 4 \quad : \text{constraint on demand of } x_1 \\ & x_1, x_2 \geq 0 \quad : \text{non-negativity constraints} \end{array}$$

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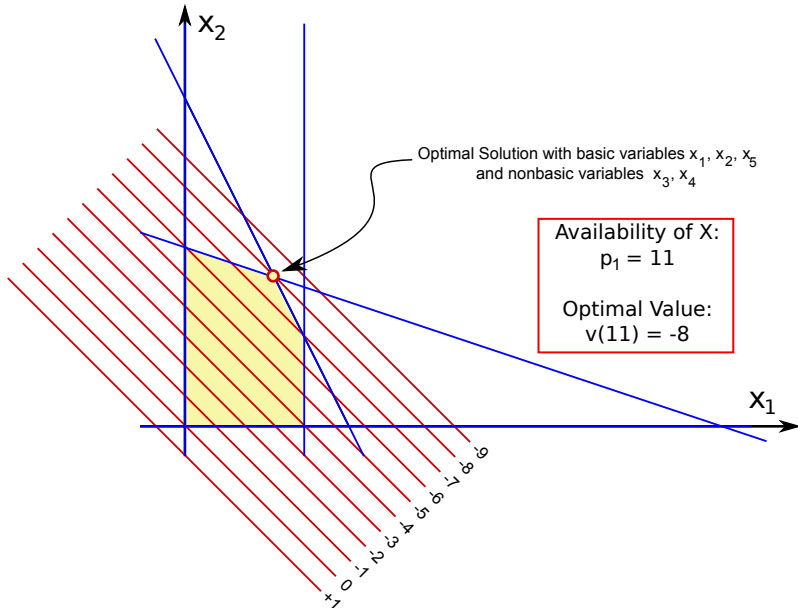
$$\begin{array}{ll} -\min & -x_1 - x_2 \\ \text{s.t.} & 2x_1 + x_2 + x_3 = p_1 \\ & x_1 + 3x_2 + x_4 = 18 \\ & x_1 + x_5 = 4 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{array}$$

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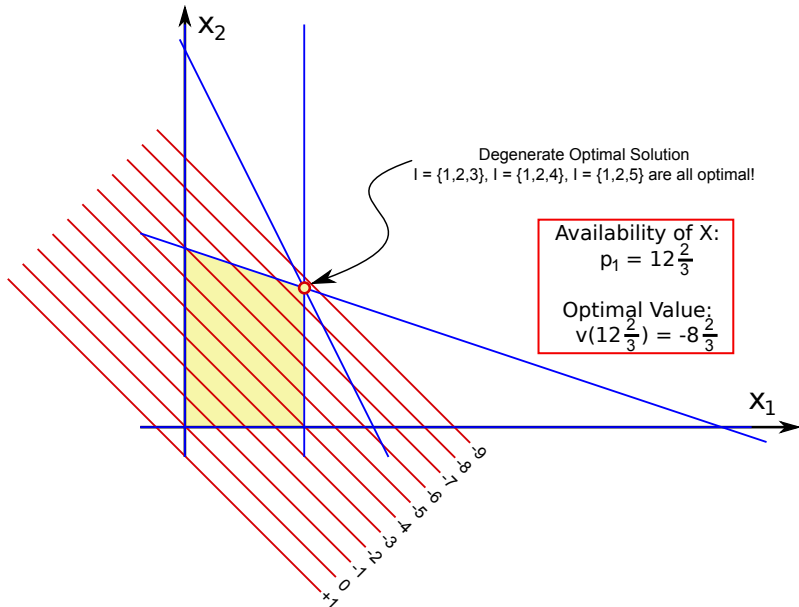
The **value function** $v(p_1)$ expresses the **optimal value** of the LP as a function of the unknown availability parameter p_1 .

$$\begin{aligned} v(p_1) = \min \quad & -x_1 - x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 = p_1 \\ & x_1 + 3x_2 + x_4 = 18 \\ & x_1 + x_5 = 4 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

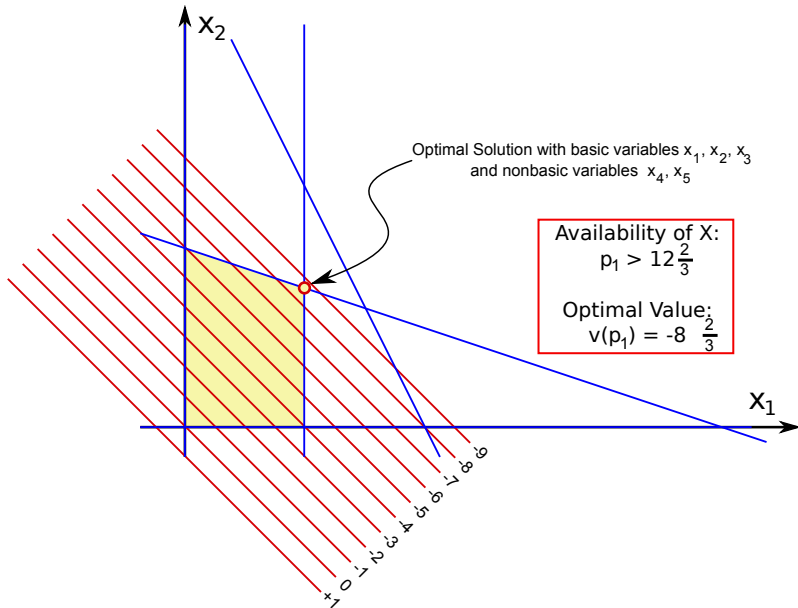
Example 1 (perturbed)



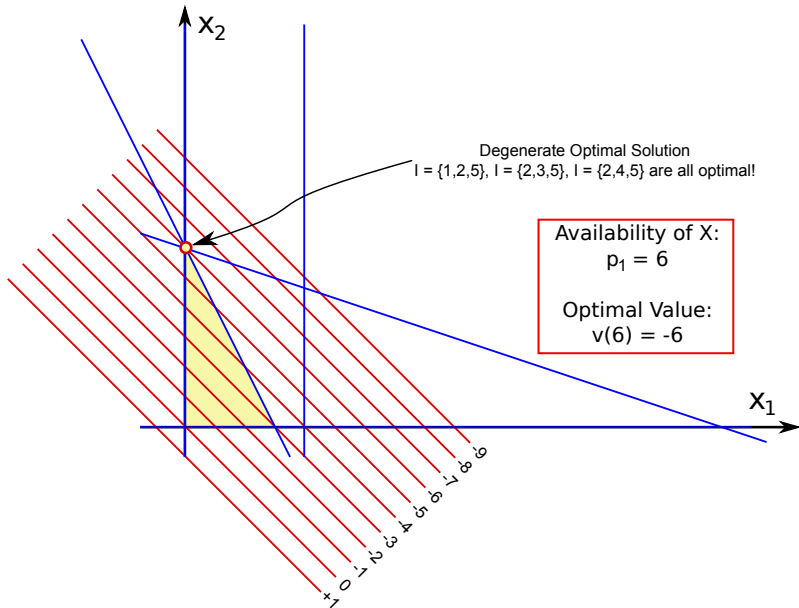
Example 1 (perturbed)



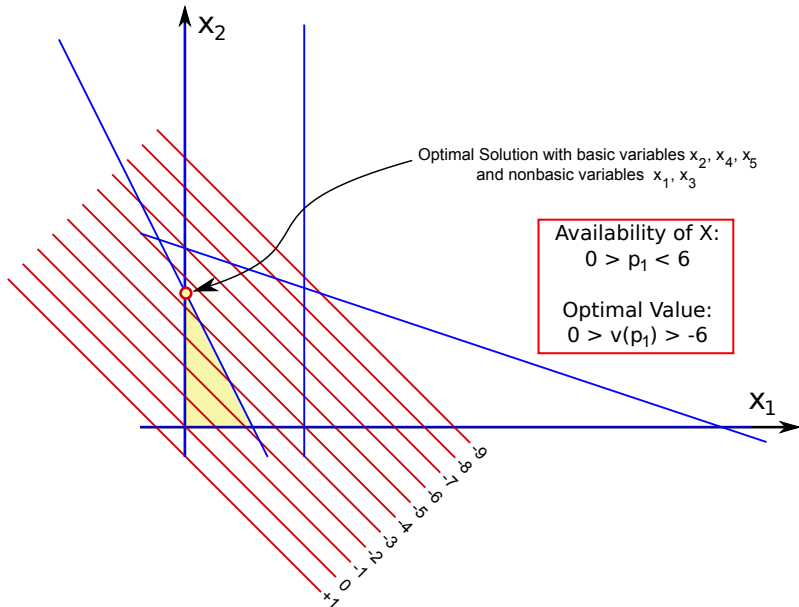
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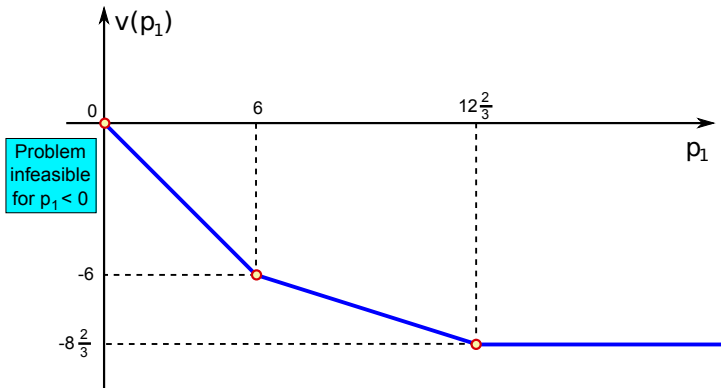


Example 1 (perturbed)



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Note: $v(p_1)$ is non-increasing, convex and piecewise linear.



Perturbation

Let $p \in \mathbb{R}^m$ denote a general RHS and define the **value function** $v(p) : \mathbb{R}^m \rightarrow \mathbb{R}$ by:

$$v(p) = \min \left\{ z = c^T x \mid A x = p; x \geq 0 \right\}$$

Solving the original LP (the **reference problem**)

$$\min \left\{ z = c^T x \mid A x = b, x \geq 0 \right\}$$

thus computes $v(b)$.

Q: what do we learn on $v(p)$ from $v(b)$?

Shadow Prices

Suppose we have solved the reference problem

$$\min \left\{ z = c^T x \mid Ax = b, x \geq 0 \right\}$$

and found an optimal basis matrix B satisfying

$$x_B = B^{-1}b \geq 0 \quad (\text{Feasibility})$$

and

$$r = c_N - N^T(B^{-1})^T c_B \geq 0 \quad (\text{Optimality}).$$

Shadow Prices (cont)

Definition: The vector of shadow prices $\Pi \in \mathbb{R}^m$ is defined as

$$\Pi = (B^{-1})^T c_B,$$

where $B = B(I)$ is an optimal basis.

Note that there can be more than one optimal basis

$\Rightarrow$ The shadow prices need not be unique.

The shadow prices give information about the sensitivity of the value function $v(p)$ at $p = b$.

Behaviour of Value Function

Theorem: $v(p) = v(b) + \Pi^T(p - b)$, $\forall p \in \mathbb{R}^m$ with $B^{-1}p \geq 0$.

Proof:

- ▶ If $B^{-1}p \geq 0$, then B remains the optimal basis for

$$\min\{z = c^T x : Ax = p, x \geq 0\}$$

since r is not affected by changing b to p .

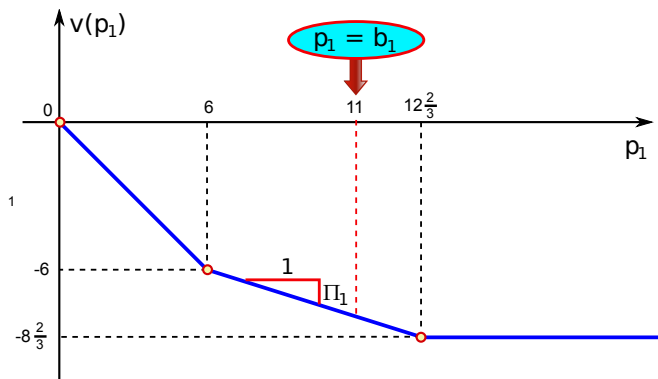
- ▶ Thus, we find

$$\begin{aligned} v(p) &= c_B^T B^{-1}p \\ &= c_B^T B^{-1}b + c_B^T B^{-1}(p - b) \\ &= v(b) + \Pi^T(p - b) \quad \square \end{aligned}$$

Theorem: $v(p) \geq v(b) + \Pi^T(p - b)$, $\forall p \in \mathbb{R}^m$.

Shadow Prices in Example 1

Note: Π_1 is the **shadow price** for the budget of **machine X**.



At $p_1 = b_1 = 11$, the **optimal costs** change by $\Pi_1 = -\frac{2}{5}$ if the **availability of X** increases by 1.

Interpretation

- ▶ Assume the company can buy a "small" additional amount of time on machine X, at price μ_1 per unit.
 - ▶ Is it worthwhile to buy additional time on X?
 - ▶ Yes if $\mu_1 + \Pi_1 < 0$ (overall cost decreases);
 - ▶ No if $\mu_1 + \Pi_1 > 0$ (overall cost increases).
- ⇒ Therefore, $-\Pi_1$ is the maximum price one should pay for one additional unit of time on machine X!

Evaluation of Shadow Prices

Q: Can we read shadow prices from final tableau?

Lemma: Suppose row t is initially a “ $\leq$ ” constraint and a **slack variable** x_s had been added. Then, $\Pi_t = \beta_s$, where β_s is the objective coefficient of x_s in the final (optimal) tableau.

Proof:

► If x_s is **nonbasic** in the final tableau, then

$$\begin{aligned}\beta_s &= -r_s = -(c_N - N^T(B^{-1})^T c_b)^T e_s \\ &= -c_s + \Pi^T a_s = 0 + \Pi^T e_t = \Pi_t.\end{aligned}$$

where e_s is a vector of zeros except for a one in the s -th position, $a_s = Ne_s$ is column s of A , and since x_s is the slack for row t we noted that $c_s = 0$ and $a_s = e_t$.

Evaluation of Shadow Prices (cont)

If x_s is **basic** in the final tableau, then

$$\beta_s = 0 = c_s = e_s^T c_B = e_s^T B^T \Pi = e_t^T \Pi = \Pi_t.$$

Which completes the proof.

Lemma: Suppose row t is initially a “ $\geq$ ” constraint and a **surplus variable** x_s had been added. Then, $\Pi_t = -\beta_s$, where β_s is the objective coefficient of x_s in the final tableau.

Example 1 (revisited)

The final tableau for Example 1 is:

BV	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	0	0	$-\frac{2}{5}$	$-\frac{1}{5}$	0	-8
x_2	0	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	0	5
x_5	0	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	1	1
x_1	0	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	3

- ▶ The constraint on the availability of X was standardised by introducing the slack variable x_3 .
- ▶ The shadow price Π_1 for that constraint thus coincides with the coefficient of x_3 in the objective row of the above tableau
 $\Rightarrow \Pi_1 = -\frac{2}{5}$.